WORKER-SAFETY COMPUTER VISION · RESEARCH PROTOTYPE

SafeSight AI

Hard-hat and restricted-zone monitoring from ordinary video, running fully offline on one workstation.

Final model M5 (YOLO26n) · application no-helmet threshold 0.60 · 2 November 2026

THE SYSTEM

From a video frame to an incident with evidence

1 · Detect

YOLO26n finds person, helmet and no_helmet in each frame.

2 · Track

ByteTrack gives each worker a temporary number. No faces, no identities.

3 · Decide

Head evidence is voted over about 0.5 s: COMPLIANT, VIOLATION or UNCERTAIN.

4 · Zones

A worker’s foot point inside a drawn polygon raises a zone alert.

5 · Record

SQLite incidents with a screenshot, cooldown and broken-track suppression.

6 · Show

A five-page dashboard, all numbers read from the database.

Offline on one workstation (RTX 2000 Ada). A separate watchdog supervised every training run.

TRAINING DATA

Three public datasets, audited and cleaned

Source	What it contributes	Licence
Construction-PPE	Worksite helmets; boxes extended to full heads for M4 and M5	AGPL-3.0
Mendeley PPE v6	Bare and helmeted heads; decisive for no_helmet	CC BY 4.0
SH17, 1,819-image subset	Bare-head labels derived from head and helmet overlap; ambiguous cases rejected	CC BY-NC-SA 4.0
Pose-teacher labels	Person boxes where datasets left people unlabelled	AGPL-3.0 model

Near-duplicates were checked against every train, validation and test split. The locked test sets were run once, after the model was chosen.

Sources and attribution: DATASETS.md. SH17 makes the final model non-commercial.

MODEL HISTORY

Six models, each answering one question

Model	Change	Common val mAP50	SH17 no_helmet AP50
B0	Construction-PPE only, leakage cleaned	0.450	0.166
M1	+ Mendeley	0.509	not run
M2	+ pose-teacher persons, masked unlabelled heads	0.703	0.513
M3	+ SHWD helmets	0.665	0.723
M4	M2 fine-tuned with full-head box style	0.652	0.439
M5	M4 fine-tuned with audited SH17 subset (final)	0.656	0.827

Validation only; AP covers all confidences, so no threshold applies. Source: reports/common_val.json.

READ THIS FIRST

M5 has two no-helmet thresholds, for two jobs

COMPARISON THRESHOLD

0.50

One rule for every model, applied to SH17 validation only: the lowest confidence that reaches 0.95 precision. It gave M2 0.65, M3 0.50 and M5 0.50.

Used for: comparing models fairly.

APPLICATION THRESHOLD

0.60

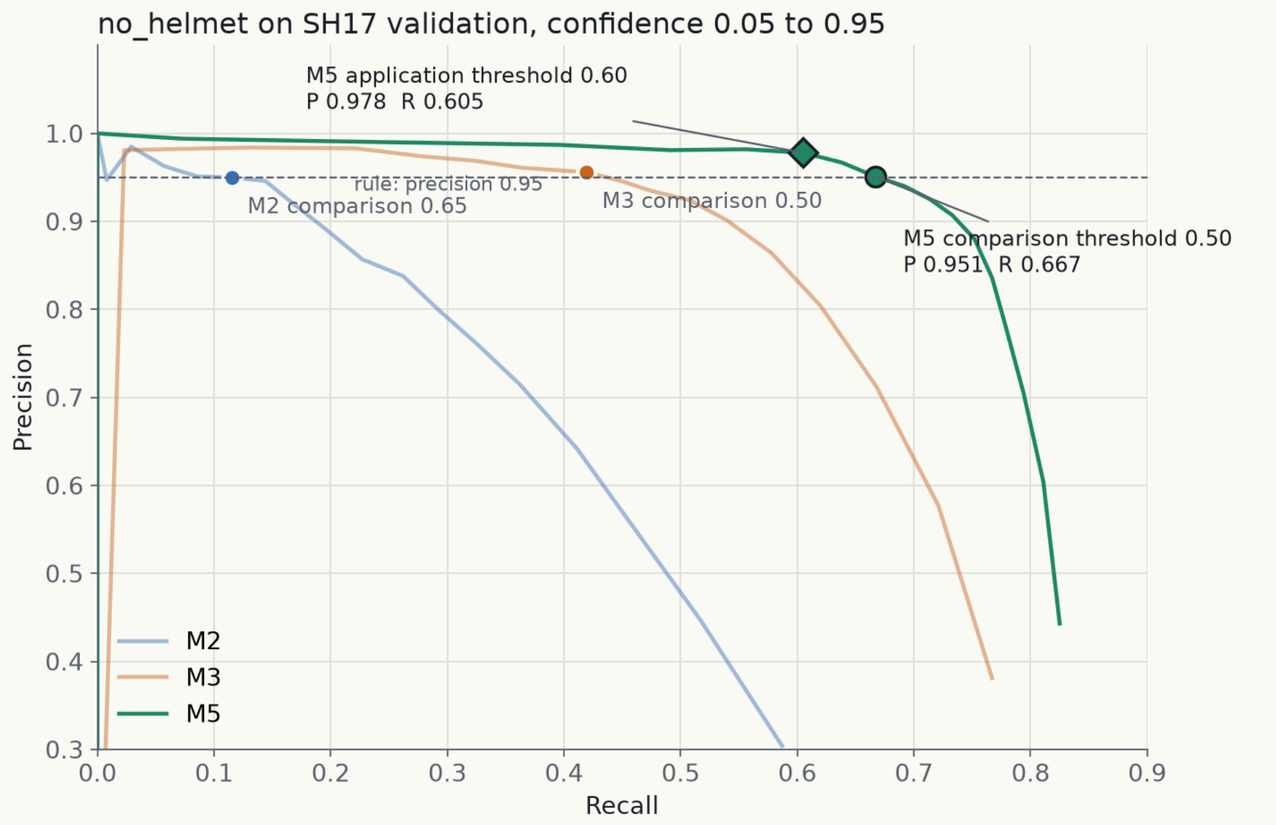
Set earlier, from validation and five demo clips. At 0.50, M5 raised 2 false alarms on those clips; at 0.60, none.

Used for: the running app and every application result.

Every result on the following slides names the threshold it used.

VALIDATION ONLY

Where both thresholds sit on M5’s curve



M5 AT 0.50, COMPARISON

Precision 0.951 · recall 0.667

M5 AT 0.60, APPLICATION

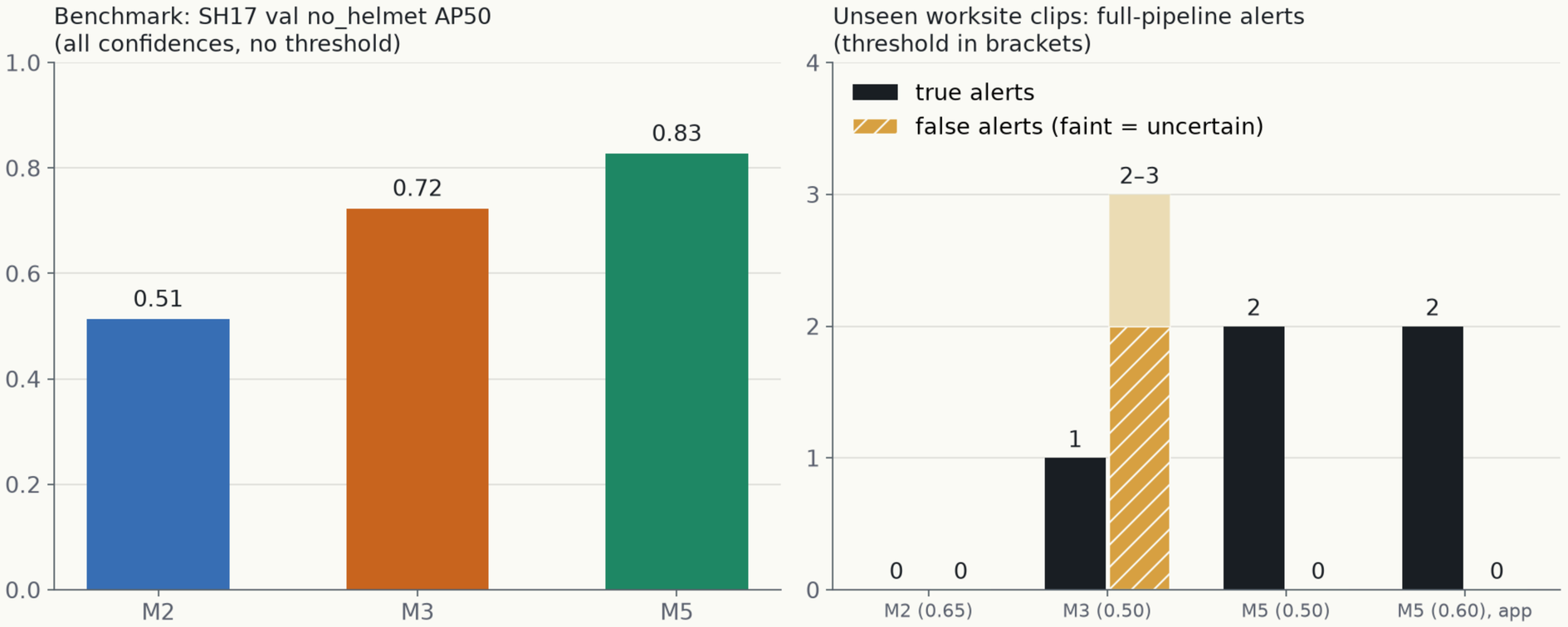
Precision 0.978 · recall 0.605

The app gives up some recall for fewer false alarms. M5’s curve is above M2’s and M3’s across the range.

SH17 validation, fully labelled. Source: reports/pr_curves.json.

CONTROLLED COMPARISON

A strong benchmark score did not guarantee clean site alerts



Left: no threshold. Right: each model at its comparison threshold, plus M5 at the application threshold 0.60.

UNSEEN FOOTAGE, FULL PIPELINE

Ten public clips, labelled by eye before any run

Measure (threshold in header)	M2 (0.65)	M3 (0.50)	M5 (0.50)	M5 (0.60) app
True no-helmet alerts, worksite clips	0	1	2	2
False alerts, worksite clips	0	2–3	0	0
Worker in soft cap, clip c01	Missed	Caught	Caught	Caught
5–7 distant bare heads, c09 and c10	All missed	All missed	All missed	All missed
False no_helmet per 100 frames, helmeted c02	0.2	10.1	0.9	0.1
Interview speaker flagged, c07 (not a site)	No	No	Yes	No

No threshold was tuned on these clips. Source: reports/unseen_compare/ and reports/unseen_gt/.

BENCHMARK VERSUS SITE

Why the two disagree

SH17 barely tests helmets

Only 152 helmets in its validation set. M3’s weakness, calling hard hats bare, hardly shows there.

Quiet is not safe

M2 at 0.65 raises no false alarms because it rarely fires at all, so it misses the real violator.

No distant workers

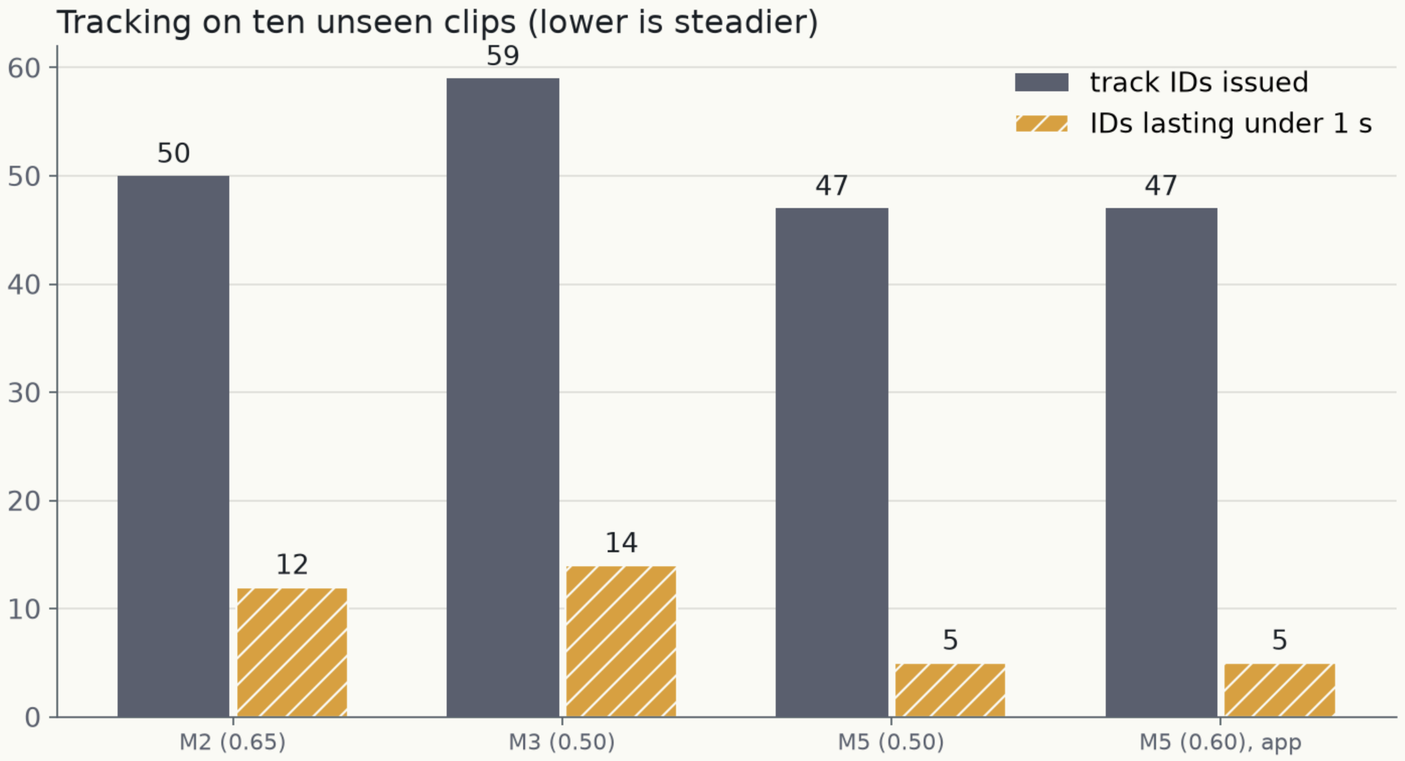
Neither benchmark has tiny, low-quality figures like c09 and c10, where every model fails.

Frames versus tracks

Benchmarks score single images. The app votes over several frames, so stray single-frame mistakes do not become alerts.

TRACKING AND SPEED

M5 keeps workers on the steadiest IDs



MEAN FPS, FULL PIPELINE

M2 62 · M3 60 · M5 58

Tracking uses person boxes only, so the no-helmet threshold does not change IDs.

Counts are tracks, not people: one worker can get two IDs after an occlusion.

Ten unseen clips, up to 1,500 frames each. Source: reports/unseen_compare/summary.json.

THE APPLICATION, VERIFIED WITH THE FROZEN MODEL

Every dashboard number comes from the database

4 of 4
true no-helmet incidents on the demo clips, 0 false, at 0.60

32
automated tests passed with the final weights

43–51
FPS in the live dashboard, detection to display

At freeze, Overview matched the database exactly: 7 sessions, 27 confirmed tracks, 10 incidents (5 no-helmet, 5 zone) and 85.7% compliance.

Application threshold 0.60 throughout. Each new demo run adds a session, so live totals will be higher.

DECISION

M5 is the final model, running at a no-helmet threshold of 0.60

Chosen on balance, not on SH17 alone: real violators caught, no false alarms on unseen site footage, steadiest tracking. M2 is kept as a read-only fallback, and M3 is preserved.

Frozen 2 November 2026. Not retrained after selection; the locked test sets were run once.

HONEST LIMITS

A research prototype, not a certified safety system

What it cannot do yet

- Never run on a real site’s cameras
- Misses small, distant workers
- No sense of context: flags any bare head
- Threshold set on little footage; a site should re-tune it
- Lower Construction-PPE test score than M2, mostly box style

Privacy and licences

- No face recognition; IDs are temporary numbers
- Runs locally; nothing leaves the machine
- Evidence images show people: consent and retention needed
- SH17 makes the model non-commercial
- Ultralytics code and weights are AGPL-3.0

LIVE DEMO, ABOUT 10 MINUTES, OFFLINE

Running order

Step	Page	Point out
1	Overview	Totals come from the database, not mock data
2	Live: helmet put on	VIOLATION, alert, then COMPLIANT; same ID throughout
3	Live: jackhammer, night	Zone alerts, a bare-headed passer-by, one UNCERTAIN
4	Live: track maintenance	Several workers, zone over the rail, helmets left alone
5	Incidents	Evidence screenshot, temporary ID, time; export CSV
6	Analytics, Configuration	Charts from records; threshold slider at 0.60

Start: .venv\Scripts\python.exe launch.py, then open localhost:8501. Full guide: DEMO_GUIDE.md.